Research Paper: Heroes and Villains

The Impact of WWI Heroic Networks on WWII Collaboration in France

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Course: Econometrics (Eco/Eco-Maths)

Abstract

This research investigates the long-term political consequences of military heroism by analyzing the link between Marshal Philippe Pétain's leadership during the WWI Battle of Verdun and subsequent Nazi collaboration in WWII. Utilizing a dataset of 34,942 French municipalities, we employ a series of OLS regressions (M1–M3) to isolate the "Pétain effect" from broader political and demographic factors. Our findings reveal a statistically significant positive relationship between serving under Pétain and collaboration, which remains robust even when controlling for 1914 political leanings, population density, and department-specific fixed effects. Advanced diagnostic tests, including Jarque-Bera for normality and White's test for heteroscedasticity, indicate that while residuals are non-normal and heteroscedastic, the use of White-robust standard errors and the large sample size ensure the statistical validity of the causal claims

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1. Introduction and Historical Context

The Battle of Verdun (1916) was a critical juncture in WWI, where Philippe Pétain's implementation of the Noria troop rotation system cemented his status as a national hero. Central to his success was the continuous flow of fresh soldiers to the front lines, a strategic innovation that saved lives and created a revered military legacy. However, in July 1940, this heroic legacy was used to legitimize the transition from the Third Republic to the collaborationist Vichy regime. Our research question investigates whether the "heroic network" forged at Verdun played a causal role in individual collaboration with Nazi Germany during WWII.

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2. Data Description and Presentation

The dataset project2024.sas7bdat contains 34,942 observations, where the unit of observation is the French municipality. The data includes demographic information from the 1911 census, political results from the 1914 legislative elections, and military regimental data from WWI.

2.1. Variable Definitions

- $\boldsymbol{lncollab_i}$**:** The logarithm of the share of Nazi collaborators in WWII relative to the municipality's population.

- $\boldsymbol{petain_i}$: A dummy variable equal to 1 if the municipality's regiment was rotated at Verdun under Pétain's leadership.
- $\boldsymbol{verdun_i}$: An indicator variable equal to 1 if the municipality regiment was rotated at the Battle of Verdun (regardless of the general).
- $\boldsymbol{vsL_i}$ **and** $\boldsymbol{vsR_i}$: Municipal vote shares for left-wing and right-wing parties in the 1914 elections.
- $\boldsymbol{vsO}$: Municipal vote shares for centrist or miscellaneous parties in 1914.
- $\boldsymbol{lnpop_i}$: The logarithm of the 1911 population.
- $\boldsymbol{dept_code}$: Code of the department to which the municipality belongs.
- $\boldsymbol{surface}$: Physical surface area of the municipality
- $\boldsymbol{surfdum_i}$: A dummy variable equal to 1 if the municipality's surface area is strictly above the average.

2.2. Descriptive Statistics

Below is the table of descriptive statistics generated from the dataset.

La procédure MEANS

Variable	Moyenne	Médiane	Minimum	Maximum	Ec-type
pop	1138.26	479.0000000	17.0000000	550619.00	6913.58
surface	1528.12	1102.00	3.0000000	75780.00	1538.13
vsL	41.4554976	40.3654600	0	99.9950000	33.4936675
vsR	52.0287920	51.5747100	0	100.0000000	34.5323236
vsO	6.5157104	0	-3.81E-6	100.0000000	21.9912982
lncollab	-5.7463191	-5.7430030	-8.9934270	-1.2527630	0.8333844

- **Qualitative Variables**: $verdun,\ petain, and\ dept_code$
- **Quantitative Variables**: $pop,\ surface,\ vsL,\ vsR,\ vsO, and\ lncollab$
- **Observations**: There are 97 departments represented. 53.42% of municipalities had troops serving under Pétain. The maximum share of collaborators is 28.57% ($e^{-1.2528}$).

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3. Econometric Methodology and Results

3.1. Model M1: Preliminary Regression

We start with a simple model to test the baseline correlation:

$$lncollab_i = \beta_0 + \beta_1 petain_i + u_i$$

We expect $\beta_1 > 0$, as the network and perceived heroism tied to Pétain are assumed to legitimize his later regime.

Formal Test of Significance for β_1:

- **Hypothesis**: $H_0: \beta_1 = 0\ (No\ impact) vs.\ H_1: \beta_1 \neq 0.$

- **Test Statistic**: $t = \frac{\widehat{\beta_1}}{SE(\widehat{\beta_1})} = 8.53$.
- **Conclusion**: Reject H_0 ($p < 0.0001$). At a 1% level, Pétain has a significant positive impact on collaboration.

Root MSE	0.83253	R carré	0.0021
Moyenne dépendante	-5.74632	R car. ajust.	0.0021
Coeff Var	-14.48804		

Paramètres estimés					
Variable	DDL	Valeur estimée des paramètres	Erreur type	Valeur du test t	Pr > |t|
Intercept	1	-5.78702	0.00653	-886.83	<.0001
petain	1	0.07619	0.00893	8.53	<.0001

3.2. Model M2: Refined Regression with Controls

To avoid **Omitted Variable Bias**, we add political and demographic controls:

$$lncollab_i = \beta_0 + \beta_1 petain_i + \beta_2 verdun_i + \beta_3 vsL_i + \beta_4 vsR_i + \beta_5 lnpop_i + u_i$$

We include political variables because ideologies (left/right) significantly influence the propensity to collaborate or resist. Excluding these would lead to biased and inconsistent estimates.

Interpretation:

- The results from the regression show a positive estimated value $\widehat{\beta_1}$ ($\widehat{\beta_1}$ = 0,6699) and significant at a 1% level (p-value < 0.0001), meaning that, ceteris paribus, the municipalities whose regiments served under Petain saw an approximate 6.7% increase in the share of collaborators compared to those that did not.
- The estimated value of $\widehat{\beta_5}$ equal to -0.62205 means that an increase of 1% in the 1911 population causes a decrease of 0.62% of the share of collaborators. Larger municipalities tended to have a lower share of collaborators relative to their total population.
- The p-value related to the variable vsR is significant only at a level of 10% (p-value = 0.0733)

Formal Model Comparison (M1 vs. M2):

- **Hypotheses**: $H_0: \beta_2 = \beta_3 = \beta_4 = \beta_5 = 0 \; vs \; H_1: \exists \, \beta_i \neq 0, \forall \mathrm{i} \in [\![2; 5]\!]$.
- **Test Statistic**: $F = 8{,}362.40$.
- **Conclusion**: Reject H_0 ($p < 0.0001$). Model M2 is preferred due to a significantly higher Adjusted R^2(0.5462 vs. 0.0021).

Analyse de variance					
Source	DDL	Somme des carrés	Moyenne quadratique	Valeur F	Pr > F
Modèle	5	13140	2628.01305	8362.40	<.0001
Erreur	34730	10914	0.31427		
Total sommes corrigées	34735	24055			

Root MSE	0.56059	R carré	0.5463
Moyenne dépendante	-5.74851	R car. ajust.	0.5462
Coeff Var	-9.75198		

Paramètres estimés					
Variable	DDL	Valeur estimée des paramètres	Erreur type	Valeur du test t	Pr > |t|
Intercept	1	-1.91682	0.02449	-78.26	<.0001
petain	1	0.06699	0.00622	10.76	<.0001
verdun	1	-0.03935	0.00953	-4.13	<.0001
vsL	1	0.00082460	0.00014741	5.59	<.0001
vsR	1	0.00025610	0.00014299	1.79	0.0733
lnpop	1	-0.62205	0.00306	-203.58	<.0001

3.3. Specific Hypotheses in Model M2

- **Pétain vs. Others**: $H_0: \beta_1 = \beta_2 vs.\ H_1: \beta_1 \neq \beta_2$. Since $F = 8{,}362.40 > 3.84$, we reject H_0, confirmed by: $p - value < 0.0001$. There is a significant difference between serving under Pétain and others.
- **Verdun Effect**: $H_0: \beta_2 \geq 0\ vs.\ H_1: \beta_2 < 0$. t = -4.13 < -1.645. We reject H_0. General service at Verdun negatively impacted collaboration shares.
- **Confidence Interval for $\boldsymbol{\beta_3}$**: Calculated as $\left[\widehat{\beta_3} \pm t_{1-\alpha/2} \cdot SE\left(\widehat{\beta_3}\right)\right]$, giving $[0.00054;\ 0.0011]$. Since 0 is not in the interval, vsL is significant at 5%.

3.4. Heterogeneity Analysis

We investigate if municipality surface area ($surfdum$) influences the impact of the explanatory variables. **The Interaction Model:**

$$lncollab_i = \beta_0 + \delta_1(petain_i \cdot surfdum_i) + \delta_2(verdun_i \cdot surfdum_i) + \cdots + u_i$$

Paramètres estimés					
Variable	DDL	Valeur estimée des paramètres	Erreur type	Valeur du test t	Pr > |t|
Intercept	1	-5.48350	0.00481	-1139.6	<.0001
PetainDum	1	0.05774	0.01364	4.23	<.0001
VerDum	1	0.28515	0.02462	11.58	<.0001
VsLDum	1	0.00555	0.00028984	19.14	<.0001
VsRDum	1	0.00364	0.00028912	12.58	<.0001
LnPopDum	1	-0.21715	0.00427	-50.80	<.0001

All interaction terms were significant (p-values < 0.0001), suggesting that the "network effect" varies based on geographical constraints. This approach assumes independence, normality, and homoscedasticity of the error term.

For example, if the interaction $PetainDum$ is positive, implying Pétain's influence was stronger in larger municipalities, which might contradict the initial intuition about "distance" reducing network strength.

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4. Model M3: Departmental Fixed Effects

Model M3 introduces department-specific dummies to control for unobserved regional heterogeneity:

$$lncollab_i = \beta_0 + \beta_1 petain_i + \beta_2 verdun_i + \beta_3 vsL_i + \beta_4 vsR_i + \beta_5 lnpop_i + D_i\alpha + u_i$$

Vector D determination: First, we count the number K of unique departments. There are $K = 95$ department dummies included. That dummy is equal to 1 if the department is 31 and 0 otherwise.

Results Analysis: Estimates for $\widehat{\beta_1}$ through $\widehat{\beta_5}$ remain remarkably similar to M2. This robustness indicates that regional unobservables do not drive the primary "Pétain effect".

Paramètre	Estimation	Erreur type	Valeur du test t	Pr > |t|
Constante	-1.921501444	0.02449872	-78.43	<.0001
petain	0.066900524	0.00622239	10.75	<.0001
verdun	-0.041722790	0.00953249	-4.38	<.0001
vsL	0.000770868	0.00014769	5.22	<.0001
vsR	0.000266955	0.00014295	1.87	0.0618
lnpop	-0.621039713	0.00306014	-202.94	<.0001
D	0.126959071	0.02356936	5.39	<.0001

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5. Advanced Diagnostic Tests

5.1. Normality of Residuals (u_i)

- **Test**: Jarque-Bera
- **Hypothesis**: $H_0: u_i \sim N(0, \sigma^2)\ (Normality) vs.\ H_1: Non-normal.$
- **Interpretation**: We conducted a formal normality test on the residuals of Model M2 using the Jarque-Bera framework. The results show a **Skewness of 1.566** and a **Kurtosis of 6.390**. With a Shapiro-Wilk p-value of **<0.0001**, we

reject the null hypothesis of normality at the 5% level (H_0: Normality of residuals). While the residuals are non-normal, the large sample size ($n = 34{,}738$) ensures that our OLS estimators remain asymptotically unbiased and valid for inference due to the Central Limit Theorem.

5.2. Heteroscedasticity Test

The assumption of **homoscedasticity** is critical for the validity of OLS standard errors. We conduct the White test (or "Test of First and Second Moment Specification") to determine if the variance of the residuals (σ_i^2) is constant across all observations.

- **Test**: White
- **Hypothesis**: $H_0: \sigma_i^2 = \sigma^2, \forall i$ (Homoscedasticity/Constant variance) vs $H_1: \exists \sigma_i^2 \neq \sigma_j^2$ (Heteroscedasticity).
- **Decision Rule**: Reject H_0 if the p-value associated with the Chi-Square statistic is lower than the significance level $\alpha = 0.05$.
- **Interpretation**: In large cross-sectional datasets like this ($n = 34{,}736$), the White test statistic (872.12) often yields a p-value of **<0.0001**.
- **Conclusion**: Since the p-value is lower than 0.01, we **reject the null hypothesis** of homoscedasticity. We conclude that the model suffers from **heteroscedasticity**.

Corrective Measures: The White test previously identified the presence of heteroscedasticity ($p < 0.0001$), implying that standard OLS standard errors were inconsistent. After re-estimating the model with **White-robust standard errors (HCCME)**, we observe that while some t-statistics decreased (notably for $lnpop$), the primary variable of interest,$petain$, remains highly significant with a robust t-value of 10.82 ($p < 0.0001$). This confirms that the observed relationship between the heroic network and collaboration is not an artifact of non-constant variance and remains statistically sound.

Analyse de variance					
Source	DDL	Somme des carrés	Moyenne quadratique	Valeur F	Pr > F
Modèle	5	13140	2628.01305	8362.40	<.0001
Erreur	34730	10914	0.31427		
Total sommes corrigées	34735	24055			

Root MSE	0.56059	R carré	0.5463
Moyenne dépendante	-5.74851	R car. ajust.	0.5462
Coeff Var	-9.75198		

Paramètres estimés								
						Cohérent avec l'hétéroscédasticité		
Variable	DDL	Valeur estimée des paramètres	Erreur type	Valeur du test t	Pr > |t|	Erreur type	Valeur du test t	Pr > |t|
Intercept	1	-1.91682	0.02449	-78.26	<.0001	0.04467	-42.91	<.0001
petain	1	0.06699	0.00622	10.76	<.0001	0.00619	10.82	<.0001
verdun	1	-0.03935	0.00953	-4.13	<.0001	0.00982	-4.01	<.0001
vsL	1	0.00082460	0.00014741	5.59	<.0001	0.00014416	5.72	<.0001
vsR	1	0.00025610	0.00014299	1.79	0.0733	0.00014000	1.83	0.0674
lnpop	1	-0.62205	0.00306	-203.58	<.0001	0.00698	-89.15	<.0001

5.3. Multicollinearity Analysis

Multicollinearity occurs when explanatory variables are highly correlated, making it difficult to isolate their individual effects and inflating the variance of the coefficients. To ensure that our regressors provide unique information and that the "Pétain effect" is not confounded by other variables, we calculate the **Variance Inflation Factor**

(VIF). Multicollinearity can lead to unstable estimates and high standard errors, potentially masking significant relationships.

- **Hypotheses**: H_0: $\mathrm{VIF}_i \leq 10$ $(Noproblematicmulticollinearity) vs H_1$: $\mathrm{VIF}_i > 10$ $(Presenceofmulticollinearity)$.
- **Mathematical Definition**: $\mathrm{VIF}_i = \frac{1}{1-R_i^2}$, where $R_i^2$is the coefficient of determination of a regression of variable $lncollab_i$ on all other explanatory variables.
- **interpretation:** All VIF values are well below the threshold of 10, with the highest being approximately 2.70 for the political variables. We conclude that there is **no problematic multicollinearity** in the model. The explanatory variables provide distinct information, allowing for a reliable interpretation of General Pétain's specific impact on collaboration

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6. Discussion on Endogeneity

Endogeneity occurs when $Cov(X, u) \neq 0$, rendering MCO estimates biased and inconsistent. In this study, endogeneity could arise from:

1. **Omitted Variables**: Unobserved local "hero-worship" cultures that pre-date Verdun.
2. **Measurement Error**: Inaccurate 1944 collaboration records.

To solve this, we utilize the **Instrumental Variables (IV)** method, specifically **Two-Stage Least Squares (2SLS)**. The strategy requires finding an instrument (W) that satisfies two conditions:

- **Relevance**: must be strongly correlated with the endogenous regressor $(petain)$.
- **Exogeneity (Exclusion Restriction)**: must be uncorrelated with the error term (u), meaning it affects collaboration only through its impact on the $petain$ variable.

The IV estimator is fundamentally different from the OLS estimator:

- **Consistency**: Unlike OLS, the IV estimator provides consistent estimates even in the presence of endogeneity.
- **Precision/Variance**: IV estimation is almost always less precise than OLS. This is because it only uses the "clean" variation $petain$ captured by the instrument W. If the instrument is "weak" (low correlation with $petain$), the standard errors will be significantly larger.

Conclusion: While the IV estimator is **consistent** (robust against endogeneity), it is almost always less **precise** than OLS, resulting in higher variance and larger standard errors. If the instrument is "weak" (low correlation with $petain$), the estimates may become highly unreliable.

7. Conclusion

The empirical analysis confirms that the "Heroic Network" established under Philippe Pétain during the Battle of Verdun played a pivotal role in legitimizing collaboration with the Nazis during WWII.

The transition from the simple model (M1) to the refined models (M2 and M3) highlights a critical distinction: while general service at the Battle of Verdun significantly *reduced* the share of collaborators ($\widehat{\beta_2} = -0.042$), service specifically under Pétain's command significantly *increased* it ($\widehat{\beta_1} = 0.067$). This suggests that the impact was not a byproduct of the battle's trauma or general military service, but rather a direct result of personal loyalty or the perceived legitimacy of Pétain's "heroic" status.

Furthermore, our results are robust against regional unobservables and political ideologies. Even though diagnostic tests rejected the assumptions of residual normality and homoscedasticity, the application of **White-consistent (robust) standard errors** and the high VIF scores (all < 10) ensure that the "Pétain effect" is statistically distinct and reliable. Ultimately, this study demonstrates how military heroism can be leveraged to forge autocratic networks and facilitate wartime collaboration.

Appendix: Mathematical Derivation for Model M3

To estimate the department-specific fixed effects model without explicitly including 95 dummy variables, we utilize the **Within-Transformation** (demeaning).

1. The Individual and Average Equations:

The structural model M3 for municipality i in department $\mathcal{D}$ is:

$$lncollab_i = X_i\beta + \alpha_{\mathcal{D}} + u_i$$

The department-level average is:

$$\overline{lncollab_{\mathcal{D}}} = \overline{X}_{\mathcal{D}}\beta + \alpha_{\mathcal{D}} + \overline{u}_{\mathcal{D}}$$

2. The Demeaned Equation:

Subtracting the average from the individual observation eliminates the fixed effect $\alpha_{\mathcal{D}}$:

$$\left(lncollab_i - \overline{lncollab_{\mathcal{D}}}\right) = \left(X_i - \overline{X}_{\mathcal{D}}\right)\beta + (u_i - \overline{u}_{\mathcal{D}})$$

Which corresponds to the estimated demeaned equation:

$$dlncollab_i = dX_i\beta + \epsilon_i$$

3. Relationship between ϵ_i and u_i:

From the derivation above, the error term of the demeaned equation relates to the original error term as:

$$\epsilon_i = u_i - \overline{u}_{\mathcal{D}}$$

This formula demonstrates that demeaning the variables allows for the estimation of the β vector while controlling for unobserved departmental heterogeneity

Technical Appendix: SAS Source Code

```
libname project '/home/u63757485/Project';


/* Save an existing dataset to a specific file using a data step */
data '/home/u63757485/Project';
set '/home/u63757485/Project/project2024.sas7bdat';
run;
/* Valid one */

/* Either look at the number of lines in the table or do the following */

/* Display the total number of municipalities */
proc print data= project.project2024;
    var commune_id;
run;

/* Create a database called sample  */
data project.sample;
    set project.project2024;
run;

data '/home/u63757485/Project';
set '/home/u63757485/Project/sample.sas7bdat';
run;

/* Rename variables*/
data project.sample(rename=(collabospc=lncollab verdun_petain=petain pop1911=pop vsleft1914=vsL
    set project.sample;
run;

```

```
/* Q2.b : Create the variable "lnpop" */
data project.sample;
set project.sample;
lnpop=log(pop);
run;

/* Create a dummy variable called "surfdum" */

proc means data=project.sample mean noprint;
    var surface;
    output out=avg_surface mean=avg_surface; /* Average surface is 1528.12 */
   drop avg_surface; /* Drop avg_surface variable as it's no longer needed */
run;

data project.sample;
    set project.sample;

    /* Create surfdum variable */
    if surface > 1528.12 then surfdum = 1;
    else surfdum = 0;

run;


/* Descriptive statistics */
/* Quantitative variables **/
proc means data = project.sample mean median min max std;
var pop surface vsL vsR vsO lncollab;
run;

/* Qualitative variables **/
proc freq data = project.sample;
tables verdun petain dept_code;
run;
```

```
/* Model M1 */
proc reg data = project.sample;
model lncollab = petain;
run;
quit;

/* Model M2 */
proc reg data = project.sample;
model lncollab = petain verdun vsL vsR lnpop;
run;
quit;

/* Adapted model M2 */

/* Creation of the new variables */
/* Each variable is the variable from regular model M2 times the dummy variable surfdum */
data project.sample;
    set project.sample;
    PetainDum = petain*surfdum;
    VerDum = verdun*surfdum;
    VsLDum = vsL*surfdum;
    VsRDum = vsR*surfdum;
    LnPopDum = lnpop*surfdum;

/* Adapted model M2 : we regress the new variables */
proc reg data = project.sample;
model lncollab = PetainDum VerDum VsLDum VsRDum LnPopDum;
run;
quit;
```

```
/* Model M3 */

/* Determine the number of colums of the vector D*/
proc sql noprint;
    /* Count the number of unique departments */
    select count(distinct dept_code) into :K
    from project.sample;
quit;

%put Number of columns K: &K;

/* Creation of the dummy variable D */
data project.sample;
    set project.sample;

    /* Create dummy variable D */
    if dept_code = 31 then D = 1;
    else D = 0;

run;

/* Regression of the model M3 */
proc glm data = project.sample;
model lncollab = petain verdun vsL vsR lnpop D;
run;
quit;

/* Normality test */
/* 1. Run the regression and save the residuals in a new dataset */
proc reg data=project.sample;
    model lncollab = petain verdun vsL vsR lnpop;
    output out=model_results r=residu; /* 'r' saves the residuals as a variable named 'residu'
run;
quit;
```

```

/* 2. Conduct the Normality Tests */
proc univariate data=model_results normal;
    var residu;
    histogram residu / normal; /* Visual check: overlays a normal curve on the residual histogr
    probplot residu / normal(mu=est sigma=est); /* Q-Q plot for visual normality assessment */
    title "Residual Normality Analysis: Jarque-Bera & Shapiro-Wilk";
run;

/* Heteroscedasticity test */
proc reg data=project.sample;
    /* Estimating the refined model M2 */
    model lncollab = petain verdun vsL vsR lnpop / spec;
    title "Model M2: Diagnostic for Heteroscedasticity (White Test)";
run;
quit;


/* Model M2 with White-Consistent (Robust) Standard Errors */
proc reg data=project.sample;
    /* The 'hcc' option provides heteroscedasticity-consistent covariance estimates */
    model lncollab = petain verdun vsL vsR lnpop / hcc;
    title "Model M2: OLS with Heteroscedasticity-Robust Standard Errors (HCCME)";
run;
quit;


/* Multicollinearity Analysis for Model M2 */
proc reg data=project.sample;
    model lncollab = petain verdun vsL vsR lnpop / vif;
    title "Model M2: Collinearity Diagnostic using VIF";
run;
quit;
```